COA Quiz - Learning Unit I and II

* Indicates required question

1. Name

2. ID

3. Computer architecture is often described as which view of the system? *

   Mark only one oval.

   - The hardware engineer's view
   - The logical view or programmer's view
   - The physical implementation view
   - The mechanical view

4. What does computer organization primarily focus on? *

   Mark only one oval.

   - The instruction set architecture
   - The way hardware components are connected and controlled
   - High-level programming languages
   - Software application design

5. Who is known as the "Father of Computers"? *

Mark only one oval.

- Blaise Pascal
- Charles Babbage
- John von Neumann
- Gordon Moore

6. What was the first programmable general-purpose electronic digital computer? *

Mark only one oval.

- Pascaline
- ENIAC
- Harvard Mark 1
- IBM System/360

7. Which technology was used in first-generation computers (1945-1954)? *

Mark only one oval.

- Transistors
- Vacuum tubes
- Integrated circuits
- Microprocessors

8. Moore's Law states that the number of transistors on a chip doubles approximately every how many years? *

Mark only one oval.

- One year
- Two years
- Five years
- Ten years

9. Which integration level refers to a chip with more than 1,000,000 transistors? *

Mark only one oval.

- SSI
- MSI
- VLSI
- ULSI

10. What is the primary role of the Arithmetic Logic Unit (ALU)? *

Mark only one oval.

- To fetch instructions from memory
- To perform calculations and logic operations
- To store permanent data
- To coordinate I/O devices

11. Which component is considered the "nerve center" of the CPU, generating control signals? *

Mark only one oval.

- ALU
- Control Unit
- RAM
- Register Set

12. What type of memory is volatile and loses its contents when power is switched off? *

Mark only one oval.

- ROM
- RAM
- Hard Disk
- Flash Memory

13. Which register holds the address of the next instruction to be executed? *

Mark only one oval.

- Instruction Register
- Program Counter
- Memory Data Register
- Stack Pointer

14. What does an opcode specify in a machine instruction? *

Mark only one oval.

- The data to be used
- The operation to be performed
- The memory address for the result
- The clock speed

15. In which stage of the instruction cycle does the CPU retrieve an instruction from memory? *

Mark only one oval.

- Execute
- Decode
- Fetch
- Store

16. Which architecture stores both instructions and data in the same memory? *

Mark only one oval.

- Harvard Architecture
- von Neumann Architecture
- RISC Architecture
- CISC Architecture

17. Which of the following is a characteristic of RISC architecture? *

Mark only one oval.

- Complex instructions
- Variable-length instructions
- Large number of general-purpose registers
- Rich addressing modes

18. What is the principle followed by a stack data structure? *

Mark only one oval.

- FIFO (First-In-First-Out)
- LIFO (Last-In-First-Out)
- LILO (Last-In-Last-Out)
- Random Access

19. Which addressing mode specifies the operand directly as a constant value within the instruction? *

Mark only one oval.

- Direct Addressing
- Immediate Addressing
- Register Addressing
- Indirect Addressing

20. What is an interrupt? *

Mark only one oval.

- A permanent error in the CPU
- A signal to the CPU for immediate attention
- A type of memory storage
- A high-speed bus

21. Which gate is considered a "Universal Gate"? *

Mark only one oval.

- AND
- OR
- NAND
- XOR

22. A Half Adder can add how many bits? *

Mark only one oval.

- One 1-bit number
- Two 1-bit numbers
- Three 1-bit numbers
- One 8-bit number

23. What is the main purpose of pipelining? *

Mark only one oval.

- To increase the clock speed
- To increase instruction throughput
- To reduce the number of registers
- To increase the size of memory

24. Which hazard occurs when instructions share resources like the same memory unit? *

Mark only one oval.

- Data hazard
- Structural hazard
- Control hazard
- Timing hazard

25. A 64-bit data bus can move how many bytes at once? *

Mark only one oval.

- 4 bytes
- 8 bytes
- 16 bytes
- 64 bytes

26. What is the duration of one clock pulse called? *

Mark only one oval.

- Instruction cycle
- T-state
- Fetch time
- Throughput

27. Which addressing mode uses a register that contains the memory address of the operand? *

Mark only one oval.

- Direct Addressing
- Register Indirect Addressing
- Immediate Addressing
- Implied Addressing

28. What happens to the Stack Pointer (SP) during a PUSH operation in a memory stack? *

Mark only one oval.

- It increases
- It decreases
- It stays the same
- It is cleared to zero

29. Which type of control unit uses fixed logic circuits like gates and decoders? *

Mark only one oval.

- Microprogrammed Control Unit
- Hardwired Control Unit
- Software Control Unit
- Virtual Control Unit

30. What is the "Speedup" of a pipelined processor? *

Mark only one oval.

- The total number of instructions
- The ratio of non-pipelined execution time to pipelined execution time
- The clock frequency in GHz
- The size of the cache

31. Which of the following is an example of an output device? *

Mark only one oval.

- Mouse
- Scanner
- Printer
- Keyboard

32. What is the binary representation of the decimal number 5? *

Mark only one oval.

- 110
- 101
- 1010
- 1100

33. Which register temporarily holds data read from or written to memory? *

Mark only one oval.

- MAR
- MBR/MDR
- PC
- IR

34. What does a compiler do? *

Mark only one oval.

- Translates assembly language to machine code
- Translates high-level language to assembly/machine code
- Executes instructions directly
- Manages I/O interrupts

35. Which adder type is generally the fastest for modern CPUs? *

Mark only one oval.

- Ripple Carry Adder
- Carry Look-Ahead Adder
- Half Adder
- Serial Adder

36. What is the primary advantage of the Harvard architecture over von Neumann? *

Mark only one oval.

- It is cheaper to build
- It allows instructions and data to be accessed in parallel
- It uses a single bus
- It is much older

37. Which logic gate produces a high (1) output only if both inputs are high? *

Mark only one oval.

- OR
- XOR
- AND
- NAND

38. What is a "nibble"? *

Mark only one oval.

- 1 bit
- 4 bits
- 8 bits
- 16 bits

39. Which register is used to point to the top of the stack in memory? *

Mark only one oval.

- Program Counter
- Instruction Register
- Stack Pointer
- Accumulator

40. What is the 2's complement of the binary number 0101 (+5)? *

Mark only one oval.

- 1010
- 1011
- 1100
- 11

41. Which component is responsible for decoding the opcode in a hardwired control unit? *

Mark only one oval.

- ALU
- Decoder
- Program Counter
- RAM

42. What is the typical range for an 8-bit unsigned number? *

Mark only one oval.

- -128 to 127
- 0 to 255
- -255 to 255
- 0 to 127

43. Which hazard occurs due to branch instructions in a pipeline? *

Mark only one oval.

- Data hazard
- Control hazard
- Structural hazard
- Timing hazard

44. What is the primary material used to make semiconductors? *

Mark only one oval.

- Copper
- Silicon
- Gold
- Plastic

45. In a Ripple Carry Adder, what causes the delay? *

Mark only one oval.

- The ALU speed
- Carry propagation through each bit
- The clock frequency
- The number of registers

46. Which addressing mode is used in instructions like NOP or HLT? *

Mark only one oval.

- Immediate
- Implied
- Direct
- Indirect

47. What is the unit for measuring CPU clock speed? *

Mark only one oval.

- MB
- GB
- GHz
- Watts

48. What is a "micro-operation"? *

Mark only one oval.

- A complex software program
- A low-level hardware action performed in one T-state
- A type of microprocessor
- A large memory unit

49. Which bus is used to identify the memory location for a read/write operation? *

Mark only one oval.

- Data Bus
- Control Bus
- Address Bus
- Expansion Bus

50. What was the main drawback of the first-generation computers? *

Mark only one oval.

- They were too small
- They dissipated a huge amount of heat
- They used transistors
- They were too fast

51. Which register stores the current instruction being decoded and executed? *

Mark only one oval.

- PC
- MAR
- IR
- MBR

52. What is the primary role of Instruction Set Architecture (ISA)? *

Mark only one oval.

- To define physical layout
- To act as interface between software and hardware
- To manage power supply
- To control fan speed

53. Which paradigm is characterized by a small set of simple, fixed-length instructions? *

Mark only one oval.

- CISC
- RISC
- MISC
- VLIW

54. What part of an instruction specifies "what to do"? *

Mark only one oval.

- Operand
- Mode
- Opcode
- Displacement

55. An 8-bit operand size is commonly referred to as a: *

Mark only one oval.

- Word
- Double word
- Quad word
- Byte

56. In the instruction cycle, which stage involves understanding the operation and identifying operands? *

Mark only one oval.

- Fetch
- Store
- Decode
- Execute

57. What is the "heartbeat" of the processor that synchronizes all operations? *

Mark only one oval.

- Power supply
- Clock signal
- BIOS
- Hard drive

58. Which CPU type contains four "mini-CPUs" inside the main chip? *

Mark only one oval.

- Single-Core
- Dual-Core
- Quad-Core
- Octa-Core

59. During the Fetch stage, which register's content is copied to the MAR? *

Mark only one oval.

- Instruction Register
- Program Counter
- Accumulator
- Status Register

60. What happens to the Program Counter (PC) immediately after an instruction is fetched? *

Mark only one oval.

- Cleared to zero
- Decremented
- Incremented
- Stored in ALU

61. Which stage is skipped if the instruction does not produce a data result to be saved? *

Mark only one oval.

- Fetch
- Decode
- Execute
- Store

62. What is an interrupt service routine (ISR)? *

Mark only one oval.

- Hardware error
- Special program to handle interrupt
- Volatile memory type
- High-speed data bus

63. Which type of interrupt is generated by external devices like a keyboard? *

Mark only one oval.

- Software Interrupt
- Hardware Interrupt
- Internal Interrupt
- Logical Interrupt

64. What does "IRQ" stand for? *

Mark only one oval.

- Internal Reset Query
- Interrupt Request
- Instruction Register Queue
- Integrated Route Quota

65. Which architecture is known for having a "small program size"? *

Mark only one oval.

- RISC
- CISC
- Harvard
- Von Neumann

66. RISC is often referred to as which type of architecture? *

Mark only one oval.

- Complex-Memory
- Load-Store
- Direct-Access
- Parallel-Bus

67. In a memory stack, what happens to the SP during a POP operation? *

Mark only one oval.

- It increases
- It decreases
- It stays the same
- It is cleared

68. Which condition occurs when you try to PUSH data onto a full stack? *

Mark only one oval.

- Stack Underflow
- Stack Empty
- Stack Overflow
- Stack Reset

69. In which addressing mode is the operand address calculated as "PC + Displacement"? *

Mark only one oval.

- Indexed
- Relative
- Direct
- Implied

70. Which addressing mode is being used in "MOV AX, [BX]"? *

Mark only one oval.

- Immediate
- Direct
- Register Indirect
- Implied

71. What is the 2's complement of binary 0011? *

Mark only one oval.

- 1100
- 1101
- 1110
- 1011

72. Which logic gate is the "NOT of XOR"? *

Mark only one oval.

- NAND
- NOR
- XNOR
- AND

73. Which adder circuit is the fundamental building block of multi-bit adders? *

Mark only one oval.

- Half Adder
- Full Adder
- Ripple Carry Adder
- Carry Save Adder

74. Which adder type computes all carries in parallel? *

Mark only one oval.

- Ripple Carry Adder
- Carry Look-Ahead Adder
- Carry Skip Adder
- Serial Adder

75. Pipelining improves performance by increasing which of the following? *

Mark only one oval.

- Instruction latency
- Clock cycle duration
- Instruction throughput
- Register size

76. If a 5-stage pipeline executes 10 instructions, how many cycles are required? *

Mark only one oval.

- 50
- 15
- 14
- 10

77. What is the formula for Speedup (S) in a pipelined processor? *

Mark only one oval.

- n / k
- n + k
- (n * k) / [n + (k - 1)]
- k / n

78. Which pipeline hazard occurs when instructions share hardware like memory? *

Mark only one oval.

- Data Hazard
- Control Hazard
- Structural Hazard
- Branch Hazard

79. Which CPU unit decides "what the Control Unit should do next"? *

Mark only one oval.

- ALU
- Instruction Register
- Next Control State Generator
- MAR

80. What are "micro-operations"? *

Mark only one oval.

- Large software apps
- Atomic hardware actions
- High-level commands
- External signals

81. Which control unit is implemented using a "Control Memory"? *

Mark only one oval.

- Hardwired
- Microprogrammed
- Logical
- Physical

82. In which addressing mode is the operand part of the instruction? *

Mark only one oval.

- Direct
- Indirect
- Immediate
- Register

83. How many T-states does a basic Fetch operation usually take? *

Mark only one oval.

- 1
- 2
- 3
- 4

84. Which architecture is faster because it is hardware-fixed? *

Mark only one oval.

- Hardwired Control Unit
- Microprogrammed Control Unit
- Software-Defined
- Virtual

85. What does the Status Register track? *

Mark only one oval.

- Next instruction
- Condition codes (Z, C, S)
- Data bus size
- Number of cores

86. In Auto Increment Addressing, when is the register incremented? *

Mark only one oval.

- Before access
- After access
- During fetch
- Never

87. Which gate produces 1 only when an odd number of inputs are high? *

Mark only one oval.

- OR
- XOR
- AND
- NOR

88. Which memory type is faster: Registers or Cache? *

Mark only one oval.

- Cache
- Registers
- Same speed
- Depends on board

89. What is the primary material used for the crystal oscillator? *

Mark only one oval.

- Silicon
- Copper
- Quartz
- Gold

90. Which type of interrupt has higher priority for system control? *

Mark only one oval.

- Hardware
- Software
- Mouse
- Printer

91. What does "LIFO" stand for? *

Mark only one oval.

- Last-In-Fast-Out
- Last-In-First-Out
- Low-Input-Fast-Output
- Logical-Index-First-Order

92. Which bus carries commands like "Read" or "Write"? *

Mark only one oval.

- Data Bus
- Address Bus
- Control Bus
- External Bus

93. What is the typical range for a 4-bit unsigned number? *

Mark only one oval.

- -8 to 7
- 0 to 15
- 0 to 255
- -7 to 8

94. Which adder is used specifically in hardware multipliers? *

Mark only one oval.

- Ripple Carry
- Full Adder
- Carry Save Adder
- Half Adder

95. In a space-time diagram, what does the horizontal axis represent? *

Mark only one oval.

- Hardware stages
- Clock cycles (time)
- Instruction size
- Memory addresses

96. Which component translates instructions into low-level signals? *

Mark only one oval.

- ALU
- Decoder
- MBR
- PC

97. What is "Pipeline Efficiency"? *

Mark only one oval.

- Total stages
- Speedup / Number of stages
- Clock frequency
- Data bus size

98. Which hazard is caused by "branch" or "jump" instructions? *

Mark only one oval.

- Structural
- Data
- Control
- Resource

99. Which architecture type is easier to modify via firmware? *

Mark only one oval.

- Hardwired
- Microprogrammed
- RISC
- ASIC

100. What is the duration of one clock pulse in a 1 GHz processor? *

Mark only one oval.

- 1 microsecond
- 1 nanosecond
- 1 millisecond
- 1 picosecond

101. Which register points to the next microinstruction? *

Mark only one oval.

- PC
- MAR
- CAR
- IR